



Document Management System



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Title: Management of Hyperkalemia in Children

Statement

This protocol is used to guide the initial management of hyperkalemia in childrenbelow age of 13 years.

Introduction

Potassiumis mainly an intracellular ion. In healthy humans the extracellular potassiumconcentration is maintained between 3.5- 5.0 mmol/L.

Hyperkalemiais defined as serum potassium above 5.5 mmol/L in children above 1 month ofage. Neonates can tolerate normal serum potassium levels upto 6.0 mmol/L.

Causesof Hyperkalemia

Spurious(False) Hyperkalemia

- Mechanicaltrauma during venipuncture/ Torniqtet/ Hemolysed sample
- HereditarySpherocytosis

TrueHyperkalemia

- **Increasedpotassium Intake/Load** (*associated with decreased renalexcretion*)
- **Increased shiftof potassium out of cells**
 - o Metabolicacidosis
 - o Insulindeficiency
 - o Increasedtissue catabolism (Rhabdomyolysis, Tumour Lysis syndrome...etc)
 - o Severe exercise

- o Renal impairment
- o High Renin/ LowAldosterone: Adrenal insufficiency(Addison disease), salt losing CAH (21Hydroxylase deficiency)
- o Pseudohypoaldosteronism

ClinicalManifestations of Hyperkalemia:

- Muscle weakness,paresthesia, hyporeflexia
- Cardiac Arrhythmias:ventricular fibrillation, asystole

ECGchanges of Hyperkalemia:

Peaked Twaves, Depressed ST segments, Decreased R wave amplitude, Prolonged PR interval,Small or absent P waves, Wide QRS complexes, Sine wave pattern. Asystole.

Investigations:

Completeblood count (CBC), Creatine Kinase (CK), Urea and electrolytes (UE1), Boneprofile and magnesium levels. In addition to serum osmolality, Renin and Aldosteronelevels. Urinary potassium and urine osmolality.

Management:

Attend patient immediately
Obtain set of vitals & brief history
If Patient is unstable attend Airway/Breathing and Circulation|
Discontinue all potassium INTAKE (oral & Intravenous)
Repeat UE1 urgently & blood gas
ECG
Connect to Cardiac monitor

v * If serum potassium <6.5 mmol/L and peaked T wave are the onlyECG changes:

- o Salbutamol2.5-5.0 mg Nebulizations, 3 doses back to back.
- o Calcium orSodium Resonium 1 g/kg PO/PR.
- o Check serumpotassium every 4 hours, if etiology is not progressive. If patient known to have renal insufficiency or previoushyperkalemia episodes then repeat in 1 hour.

v * If serum potassium 6.5-7.1 mmol/L and peaked T wave is only ECGchanges:

- o Salbutamol2.5-5.0 mg Nebulizations, 3 doses back to back. *Monitor for tachycardia andfine tremor.*
- o 8.4% SodiumBicarbonate 1-2 mmol/kg infuse over 30 minutes. *Monitor ionized calciumlevels.*
- o Calcium orSodium Resonium 1 g/kg PO/PR. *Monitor for hypokalmemia.*
- o Dextrose **AND**Insulin (Monitor blood sugar): *Monitor for hypoglycemia.*

§ **Bolus:**Give D10W 5-10 ml/kg **and** Insulin 0.1-0.2 U/kg IV over 30 minutes.

§ **Continuous:** D10W5 mL/kg/hr **and** Insulin 0.1 unit/kg/hr

- o Frusemide 1mg/kg iv to promote potassium excretion through urine (if BP not low and childnot dehydrated). *Monitor for dehydration.*

- o Hemodialysis if above measures fail or if patient is in renal failure.

v *** If serum potassium ABOVE 7.1 mmol/L or there are ECG changes other than peaked Twaves:-**

- o Salbutamol 2.5-5.0 mg Nebulizations, 3 doses back to back. *Monitor for tachycardia and fine tremor.*
- o Calcium Gluconate 10% for cardiac muscle stabilization: 0.5 mL/kg, iv over 5-10 minutes.
- o 8.4% Sodium Bicarbonate 1-2 mmol/kg infuse over 30 minutes. *Monitor ionized calcium levels.*
- o Calcium or Sodium Resonium 1 g/kg PO/PR. *Monitor for hypokalemia.*
- o Dextrose **AND** Insulin (Monitor blood sugar): *Monitor for hypoglycemia.*

§ **Bolus:** Give D10W 5-10 mL/kg **and** Insulin 0.1-0.2 U/kg IV over 30 minutes.

§ **Continuous:** D10W 5 mL/kg/hr **and** Insulin 0.1 unit/kg/hr

- o Furosemide 1 mg/kg iv to promote potassium excretion through urine (if BP not low and child not dehydrated). *Monitor for dehydration.*
- o Recheck serum potassium Every hour until levels are below 6.5 mmol/L
- o Hemodialysis if above measures fail or if patient is in renal failure.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
The HSC Handbook of Pediatrics. 10 th edition.	Dam Chend, Brent a Williams	2003	235-237
Comprehensive Pediatric Nephrology	Denis F Geary, Franz Schaefer	2008	407-410

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